

TATALINA, AK, USA

WMO: 702315

Lat: 62.894N

Lon: 155.976W

Elev: 964

StdP: 14.19

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26536

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-22.7	-35.0	0.8	-27.3	-29.7	1.1	-22.4	19.3	20.4	16.7	16.2	1.5	60	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	76.2	58.7	72.8	57.1	69.5	55.5	61.1	72.0	59.2	68.8	57.5	65.8	5.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.0	71.7	63.0	55.2	67.2	60.9	53.7	63.7	59.8	27.5	72.0	26.2	69.0	25.1	66.3	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.2	11.8	DB	-30.3	80.7	5.8	2.7	-34.5	82.7	-37.9	84.2	-41.1	85.7	-45.4	87.7	(4)
(5)			WB	-30.3	63.3	4.9	2.8	-33.8	65.3	-36.7	67.0	-39.4	68.6	-43.0	70.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.8	0.8	11.1	15.9	33.3	48.0	56.6	58.8	54.3	45.1	28.2	10.8	5.9		
	DBStd	23.18	17.10	13.61	11.77	10.44	7.91	6.23	5.68	5.94	6.98	10.13	12.43	13.79		
	HDD50	7732	1525	1089	1057	503	129	12	2	22	172	676	1177	1368		
	HDD65	12505	1990	1509	1522	952	529	262	208	336	596	1141	1627	1833		
	CDD50	734	0	0	0	1	66	210	274	156	26	1	0	0		
	CDD65	31	0	0	0	0	1	10	15	5	0	0	0	0		
	CDH74	197	0	0	0	0	10	67	91	29	0	0	0	0		
	CDH80	19	0	0	0	0	0	10	7	2	0	0	0	0		
	WSAvg	4.3	3.9	4.4	4.9	5.2	5.0	4.6	4.4	4.1	4.5	4.0	3.8	3.5		
	PrecAvg	17.4	1.0	0.7	0.6	0.6	0.8	1.8	2.8	2.9	2.2	1.5	1.3	1.2		
	PrecMax	25.1	3.5	3.5	2.7	2.0	2.0	3.3	5.6	5.4	5.7	4.7	4.0	4.9		
(15) Precipitation	PrecMin	9.9	0.0	0.0	0.0	0.0	0.0	0.4	0.6	1.0	0.4	0.3	0.0	0.1		
	PrecStd	3.5	0.9	0.8	0.7	0.5	0.5	0.8	1.2	1.0	1.5	0.9	1.1	1.0		
	DB	40.7	40.1	43.5	59.4	75.5	80.3	80.4	78.0	65.6	53.1	39.3	40.1			
	MCWB	35.7	34.5	34.4	44.1	53.0	59.2	61.2	62.4	53.0	46.5	36.8	30.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	35.0	36.3	38.5	54.0	69.7	75.3	76.6	72.2	60.7	46.7	35.6	34.8			
	2% MCWB	32.1	32.8	32.5	40.9	51.5	57.0	59.8	59.4	50.5	41.9	33.7	31.9			
	5% DB	30.6	33.1	35.2	50.7	65.4	71.8	73.0	67.5	57.4	43.8	32.2	30.1			
	5% MCWB	29.0	30.9	30.5	39.3	49.4	55.3	58.6	57.0	49.6	40.3	31.1	28.7			
	10% DB	25.7	29.3	32.0	47.2	61.3	68.5	69.5	63.8	54.5	41.1	27.7	25.3			
	10% MCWB	24.5	27.4	28.5	37.4	47.5	53.7	57.2	55.1	48.5	38.4	26.9	24.0			
	0.4% WB	36.3	36.1	36.5	45.2	54.8	63.0	63.7	64.3	54.6	47.0	37.1	34.8			
	0.4% MCDB	39.5	38.2	40.7	57.2	70.1	78.2	76.4	74.9	60.6	51.7	38.5	40.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	32.4	33.3	33.5	42.0	52.9	59.1	61.2	60.8	52.5	43.1	33.4	32.0			
	2% MCDB	34.3	35.6	37.3	51.6	66.8	70.8	73.2	69.5	57.5	45.6	35.0	33.7			
	5% WB	28.8	30.5	31.3	40.0	50.7	56.9	59.6	58.3	50.9	41.1	31.1	28.6			
	5% MCDB	30.6	33.1	34.5	48.8	63.0	68.3	69.8	65.2	55.9	43.4	32.5	30.8			
	10% WB	24.4	27.3	28.4	38.2	48.7	55.1	58.3	56.3	49.5	38.5	26.9	24.2			
	10% MCDB	25.8	29.3	32.0	46.3	59.1	65.8	67.3	62.0	53.9	40.8	27.6	25.3			
	MDBR	9.2	11.0	14.7	15.5	16.1	15.7	13.8	12.2	11.0	8.4	8.7	9.4			
	5% DB	10.9	11.4	14.3	18.1	21.2	19.5	19.1	18.1	14.7	10.3	9.0	11.8			
(36) Mean Daily Temperature Range	MCWBR	9.2	9.8	10.0	9.4	9.6	8.1	8.2	8.1	8.0	7.0	8.2	9.9			
	MCDBR	11.1	11.5	12.7	16.6	19.0	17.5	16.8	15.2	12.0	9.3	9.1	11.4			
	MCWBR	9.7	10.0	9.6	9.0	9.0	8.0	7.9	7.6	7.3	6.9	8.4	9.8			
	MCDBR	9.7	10.0	9.6	9.0	9.0	8.0	7.9	7.6	7.3	6.9	8.4	9.8			
(40) Clear-Sky Solar Irradiance	taub	0.194	0.253	0.260	0.306	0.341	0.358	0.356	0.360	0.302	0.275	0.204	0.180			
	taud	2.035	2.231	2.289	2.253	2.411	2.381	2.434	2.436	2.578	2.466	2.034	1.916			
	Ebn at Noon	152	225	273	279	278	274	271	259	259	220	138	81			
	Edh at Noon	10	19	26	34	32	34	31	29	20	15	9	5			
(44) All-Sky Solar Radiation	RadAvg	82	309	886	1461	1649	1676	1436	1097	726	313	107	35			
	RadStd	10	33	86	114	172	124	123	113	100	38	17	3			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+1.80	N/A	+1.53	N/A	N/A	N/A	-538	-644	N/A	N/A

Nomenclature: See separate page